

Pavan Kumar Ambekar

Email: pavankumar9219@gmail.com **LinkedIn:** [pavankumarambekar](#) **Marital Status:** Married
Phone: (+91)9008009219 , (+60)11 5358 3184 **Address:** No.102, 8th Cross, 2nd Main, Jayanagar,
Mata Anapurneshwari, Tumkur, India-572102.

Work Experience Summary

Having a total experience of 5+ years.

- Senior DFT Engineer at "Quest-Global", Penang Malaysia and India. June 2025 to current.
- Senior DFT and Synthesis Engineer at "Mirafr Technology Pvt Ltd", Bangalore: June 2023 to June 2025
- DFT Engineer at "Cerium Systems TechMahindra", Bangalore: April 2022 to June 2023
- Contract worker for TechMahindra (Overlapped): July 2021 to March 2022
- DFT Engineer at "Ziotron Consulting Pvt Ltd", Bangalore: Jan 2020 to Sep 2020
- Trained as DFT Engineer in Sandeepani Institute of VLSI: Oct 2018 to June 2019

Technical Skills

- Spyglass DRC cleaning
- Scan Insertion & EDT Insertion.
- ATPG for S@ and TDF.
- Coverage analysis & improvement.
- Simulation (No-timing and Timing).
- MBIST insertion and Validation (GLS Simulation).
- IEEE 1149.1 and IEEE 1149.6.
- Turnkey project.
- Scripting in bash, shell & html(GUI).
- Time Management.
- Communication.
- DFT fundamentals.
- ASIC flow.
- Block level and SoC level handling.
- IP validation (DTS & PLL).

Technical EDA Tools

- DFT Design: Siemen Tessent & Cadence Genus
- LEC: Formal Compiler
- Debug method: RTL & Gate Level Simulation
- Formal Verification: VCS
- Simulation Verification: DVE, Verdi & QuestaSim
- Programming languages: Verilog, VHDL, PDL, bash, shell and html.

Experience

1. **Socionext, Japan (via Quest-Global)**
Designation: Senior DFT Engineer

Project: SHN-SNI (June 2025 to current)

Team Size: 2

Technology node: 5nm

Responsibilities:

- Cleaning DRC's at early RTL via spyglass.
- Conducting Synthesis and Scan procedures for the project.
- Implementing ATPG for Stuck-at and Transition Faults.
- Analyzing and improving test coverage to 99.6 and 96 for stuck-at and at-speed respectively.
- Performing simulation both without timing.
- Merging ex-text, in-test, and Ram-Seq statistics.
- Running Quality checks such as LEC, MUSE (erc-ccf) and DFTV

2. MediaTek (via Mirafra Technology) iD:ot906764

Designation: Senior DFT and Synthesis Engineer

Project: Youshan, Davince and Monet (Oct 2023 to June 2025)

Team Size: 10

Technology node: 2nm

Responsibilities:

- Conducting Synthesis and Scan procedures for the project.
- Implementing ATPG for Stuck-at and Transition Faults.
- Analyzing and improving test coverage to 99.6 and 96 for stuck-at and at-speed respectively.
- Performing simulation both without timing.
- Merging ex-text, in-test, and Ram-Seq statistics.
- Running Quality checks such as LEC, MUSE (erc-ccf) and DFTV

3. Mirafra Technology

Designation: Senior DFT Engineer

Project: Ariena Chip (Jun 2023 — Sep 2023)

Team Size: 1

Technology node: 5nm

Responsibilities:

- Managing the DFT process for the Ariena Chip project.
- Coordinating MBIST insertion and validation tasks.
- Overseeing scan insertion and graybox generation.
- Analyzing and enhancing scan coverage.
- Conducting pre-layout and post-layout ATPG on stuck-at and transition.

4. CDAC (Government of India), Trivandrum (via TechMahindra Cerium Systems)

Designation: DFT Engineer

Project: CDAC Vega Quad-core Chip (Mar 2023 — Jun 2023)

Team Size: 1

Technology node: 14nm

Responsibilities:

- Handling DFT procedures for the CDAC Vega Quad-core Chip project.
- Conducting MBIST and ICLNetwork tests.
- Generating and testing IJTAG-graybox.
- Performing ICLNetwork and MBIST testing from the SoC level.
- Utilizing custom PDL flow for testing ICLNetwork and IP.

5. GUC asic, Taiwan (via TechMahindra Cerium Systems)

Designation: DFT Engineer

Project: Agamotto (Nov 2022 — Feb 2023)

Team Size: 1

Technology node: 7nm

Responsibilities:

-
- Test DFT Flow based on in-house WFM data system.
6. **Intel, US (via TechMahindra Cerium Systems)** WWID:12046014
Designation: DFT Engineer
Project: INTEL Xeon Server chip (Willow-Wood XNE.WLW.DFT.COE) (April 2022 — Oct 2022)
Team Size: 5
Technology node: 7nm
Responsibilities:
- MBIST & ICLNetwork tests, Generation of IJTAG-graybox.
 - Generating & Testing ICLNetwork and MBIST from SoC level.
 - Using custom PDL flow for testing ICLNetwork and IP.
 - Good knowledge of Intel security policy in the IC (Integrated Circuit).
 - SDF/Timing Simulation.
7. **TechMahindra, (via Ziotron Consulting Pvt Ltd)**
Designation: DFT Engineer (Dec 2020 — April 2022)
Technology node: 7nm
Responsibilities:
- Handled 4 blocks of the project.
 - Scan insertion.
 - Cleaning DRC's.
 - ATPG for Stuck-at & At-speed.
 - Coverage analysis and improvements.
 - Achieved 99.1% for Stuck-at and 89% for At-speed. Zero delay simulation.
8. **Sandeepani under CoreEL Technologies**
Designation: DFT Trainee Engineer (Oct 2018 — Jun 2019)
Technology node: 28nm
Responsibilities:
- Scan Insertion.
 - Cleaning DRC's.
 - ATPG for Stuck-at & At-speed.
 - Coverage analysis and improvements.
 - No Timing and Timing simulation.
 - RTL coding.

Qualifications

- Master of Science (Microelectronics)
Newcastle University, Newcastle Upon Tyne, United Kingdom. - First Class with merit - 2020
- Bachelor of Engineering (Electronics & Communications)
Siddaganga Institute of Technology under VTU, Tumkur, India. - First Class - 2018